

Detecting Malicious Network Behaviour from Low-Level Traffic Signals

1. Context of the Problem

Modern organizations are increasingly defined by their reliance on digital networks, which generate large volumes of low-level traffic data. This project is situated in the problem domain of cybersecurity, specifically focusing on network intrusion detection. Historically, the field has relied heavily on signature-based systems that identify known patterns of malicious code. However, as cyber threats evolve, these traditional methods often struggle to detect novel, "zero-day" vulnerabilities.

A useful alternative is to view network traffic not only as discrete packets of data, but also as a signal that can be analysed statistically over time. Within this context, measurements such as packet size, duration, protocol usage, flow counts, and arrival timing constitute the measurable variables of the signal. Under normal conditions, these signals follow predictable patterns; however, malicious activity, such as Distributed Denial-of-Service (DDoS) attacks or unauthorized endpoint access, manifests as measurable anomalies. This makes cybersecurity a particularly suitable domain for Signal Processing for Statistics and Data Science (SPSDS). Just as signal processing techniques are used to isolate irregularities in an electrocardiogram or an audio wave, these same principles can be applied to network traffic data to distinguish malicious activity from the baseline of normal operations.

2. Description of the Problem

The practical problem addressed by this project is the difficulty of detecting suspicious network behaviour accurately and efficiently. Manual monitoring is far too slow for modern network environments, where large amounts of traffic are generated continuously and patterns can change within seconds. Rule-based systems are still useful, but they are limited because they depend on predefined attack signatures and may fail when malicious behaviour evolves or closely resembles legitimate activity. This is especially challenging in noisy network environments, where normal actions such as video streaming, bulk data transfers, or temporary spikes in requests can look similar to actual attacks. As a result, detection is not simply a matter of identifying anomalies, but of distinguishing harmful behaviour from harmless variation without producing too many false alarms. Attackers also adapt their methods over time, which makes static

detection approaches less reliable. For this reason, the project explores how statistical methods and machine learning can support the detection process by identifying patterns that separate malicious traffic from normal network behaviour while maintaining a practical balance between speed and accuracy (Buczak and Guven, 2015).

3. Dataset and Data Constraints

The practical part of the project will use an open intrusion detection dataset, with CICIDS2017 as the preferred option (Sharafaldin et al., 2018). A secondary candidate is KDD Cup 1999, which can serve as a fallback or comparison dataset, as it is also widely used in intrusion detection research. These datasets contain labelled examples of both benign and malicious traffic and provide structured numerical features that are suitable for classification and statistical analysis. This makes them useful for building and evaluating models that distinguish normal activity from attack behaviour. At the same time, such datasets also present limitations, including class imbalance, noisy or redundant features, and, in the case of older datasets, traffic patterns that may not fully reflect current real-world environments. These constraints need to be considered carefully when interpreting the results.

4. Possible Technologies

Several technologies could be used to address this problem. Python is an appropriate implementation environment because it supports preprocessing, modelling, and visualisation within a single workflow. Libraries such as **pandas** and **NumPy** can be used for data cleaning and handling, while **matplotlib** can support visual exploration of patterns in the traffic data. For modelling, **scikit-learn** provides suitable tools (Pedregosa et al., 2011) for methods such as Logistic Regression, Random Forest, and anomaly detection techniques. In addition, **scipy** could be used for optional signal-processing steps such as smoothing or Fourier-based exploration.

5. Proposed Technologies and Justification

This project will use Python, together with libraries such as **pandas**, **scipy**, and **scikit-learn**, to analyse the CICIDS2017 intrusion detection dataset. The methodology follows standard data science stages, beginning with exploratory analysis to understand the

data structure, identify variable types, inspect correlations, and clean the dataset where necessary. At its core, the task is a classification problem in which network traffic features, such as flow duration, timing, packet length, and related signal-based measurements, are used to predict whether traffic is benign or malicious.

The proposed analytical approach compares Logistic Regression with Random Forest on the same processed dataset. Logistic Regression is chosen as a simple and interpretable baseline model, while Random Forest is selected as the main model because it can capture more complex and non-linear relationships that may exist in cybersecurity data (Farnaaz and Jabbar, 2016). This comparison allows for a fair and measurable test of whether the more flexible model offers a meaningful improvement in detection performance. In addition, optional signal-processing methods, such as smoothing or Fourier-based exploration of packet timing, may be used for feature engineering if supported by the dataset structure. The results will be presented through appropriate visualisations to highlight differences between normal and malicious traffic patterns.

6. Relevance and Measuring Success

The project is relevant both academically and practically. From a business perspective, more accurate detection of malicious traffic could support earlier incident response, reduce operational disruption, and lower the overall risk associated with cyber attacks. In environments where networks carry large volumes of data, even small improvements in detection quality can have significant practical value. From an academic perspective, the project brings together signal-oriented reasoning, descriptive statistics, machine learning, and critical evaluation within a real cybersecurity use case. Success will be measured using accuracy, precision, recall, F1-score, false positive rate, and confusion matrix analysis. Among these metrics, false positive rate is especially important, because a system that produces too many false alarms quickly becomes less useful in practice and can overwhelm administrators. For that reason, the evaluation will not focus only on overall accuracy, but also on whether the model can detect malicious behaviour reliably without creating unnecessary alert fatigue.

Research Question:

Can low-level network traffic signals be used to detect malicious behaviour effectively, and does Random Forest outperform Logistic Regression in distinguishing malicious traffic from normal network activity?

Hypothesis:

A Random Forest classifier trained on low-level network traffic features will detect malicious behaviour more effectively than Logistic Regression, particularly in terms of F1-score and false positive rate. This advantage may increase further if signal-processing features, such as Fourier-based representations of packet timing, can be extracted from the dataset and included alongside the original traffic variables.

References

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